Bishal Lamichhane

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Summary

Backend-focused software engineer with 2+ years designing and building scalable RESTful APIs, multi-tenant SaaS architectures, and high-performance backend systems in Laravel and PHP. Migrated a legacy ERP to a modern Laravel architecture, cutting technical debt by 30% and improving maintainability by 40%. Deep expertise in domain-driven modular monoliths, the Service-Repository pattern, OAuth2 authentication, database design and query optimization (achieving up to 33% performance gains), and queue-based / real-time processing.

Technical Skills

Backend & Arch.	RESTful APIs, Service-Repository Pattern with Interface-based DI, Domain-Driven Design, Modular Monolith, Multi-Tenant SaaS Architecture, Feature-Flag Architecture, Microservices
Languages	PHP, SQL, JavaScript, TypeScript, Python
Frameworks	Laravel 12, Filament PHP, Inertia.js, Next.js, React
Databases	PostgreSQL (JSONB, Window Functions, Full-text Search, pg_trgm), MySQL, MariaDB — Query Optimization, Indexing, Schema Design
Auth & Security	OAuth2 (Laravel Passport), RBAC (Spatie Permission), httpOnly Cookies, Rate Limiting, Request ID Middleware, Activity Logging
Real-Time	WebSockets (Laravel Reverb / Echo), Queue-based Processing (Laravel Horizon + Redis)
DevOps & Tools	Docker, Redis, Git, GitLab, CI/CD Pipelines, Postman
Frontend	Next.js (App Router), React, TypeScript, Tailwind CSS (working knowledge)

Professional Experience

Software Engineer Nov 2024 – Present
Diagonal Technologies Pvt. Ltd. *Kathmandu, Nepal*

- Migrated a legacy PHP ERP to Laravel, reducing technical debt by 30% and improving maintainability by 40%
- Designed RESTful APIs using the Service-Repository pattern with interface-based dependency injection, increasing code reusability by 40%
- Optimized SQL queries and database schemas, achieving a 33% performance improvement across all system modules
- Implemented an RBAC system with secure authentication and role-scoped access, protecting sensitive enterprise data

Backend Developer Mar 2023 – Nov 2023
Red Tail Fox Pvt. Ltd. *Kathmandu, Nepal*

- Developed production APIs for an event platform serving 1,000+ concurrent users, improving response times by 12%
- Implemented a microservice architecture enabling independent service scaling and zero-downtime deployments
- Established CI/CD pipelines, reducing production bugs by 10% and accelerating deployment cycles

Backend Developer Intern Nov 2022 – Feb 2023
Red Tail Fox Pvt. Ltd. *Kathmandu, Nepal*

- Built scalable APIs with complex business logic supporting a multi-relational database architecture
- Enhanced system reliability through comprehensive error handling, reducing server crashes and debugging time

Key Projects

Enterprise ERP System | *Laravel, PHP, MariaDB, Docker*

- Modernized a legacy ERP, achieving a 40% maintainability improvement and a 33% database performance gain
- Designed RESTful APIs improving efficiency by 40%; integrated third-party APIs to reduce manual errors
- Automated workflows with job scheduling for dynamic email delivery and task execution

Learning Management System (LMS) | *Laravel, Blade, MySQL*

- Built a full-stack LMS with reusable UI components for course management, enrollment, and content delivery
- Developed a mock-test module enabling timed assessments with instant scoring and feedback
- Implemented a premium subscription and payment flow granting access to paid courses and mock tests

- Built a course review and rating system to collect student feedback and improve content quality

Multi-Tenant HMS | *Laravel 12, PostgreSQL, Reverb, Redis, Docker, Next.js*

- Architected a full-stack multi-tenant platform covering 22 feature-flag-driven modules (Reservations, Housekeeping, Billing, HR, Analytics, CRM, and more) — flags toggled per property with no redeployment
- Applied a domain-driven modular monolith with a strict Service-Repository pattern and interface-based DI; 22 independent domains, each with its own migrations, providers, and routes
- Designed a PostgreSQL schema with JSONB translatable fields, property-scoped Global Scopes for row-level multi-tenancy, and window functions for RevPAR/ADR analytics
- Built an OAuth2 auth flow (Laravel Passport) with property-scoped Spatie roles and httpOnly-cookie token storage

Work Event Management System | *Filament PHP, MariaDB*

- Architected a calendar system with recurring event patterns, real-time updates, and role-based permissions

Ramailo Games Platform | *Laravel, MySQL, Microservices*

- Optimized database queries, reducing load time from 5s to 1s (80% improvement) and boosting user engagement
- Implemented microservices with end-to-end encryption ensuring secure data handling and regulatory compliance

Real-Time Chat Application | *Laravel Reverb, WebSocket, MySQL, Inertia.js*

- Built a production-ready messaging system with authentication, group chats, typing indicators, and presence detection

Email/SMS Spam Detection | *Python, Naive Bayes, Google Colab*

- Developed an ML model achieving 95% accuracy using Naive Bayes with a comprehensive data preprocessing pipeline

Education

BSc. Computer Science and Information Technology Jul 2018 – May 2023
Tribhuvan University, National College of Computer Studies *Kathmandu, Nepal*
GPA: 82.29%

Certifications

Introduction to Generative AI — Google (March 2024)